

Coding with Large Language Models (LLMs)

May 28, 2025

IMET Workshop

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Prompt Engineering Primer

- **Prompt engineering** is the process of designing and optimizing the language input to effectively guide a LLM response.
- Effective prompt engineering requires **Context** and **Expectations** and can include a **Persona (Role)**
- In the same way you would communicate with a person, the more **context** and **detailed expectations** you provide the better the output will be from the LLM.
 - Think about walking into an ER to for triage

Do your ChatGPT prompts stink?

Give ChatGPT a role

“You are a...”

- ✓ Writing coach
- ✓ GitHub Actions expert
- ✓ C# code reviewer
- ✓ Copywriter
- ✓ {{ POSITION }} interviewer”

Chain your prompts

“Create a guide to AI. Give a headline and teaser.” [Output]

“Now create 5 subtopics.” [Output]

“Now add 3 AI techniques in each subtopic.” [Output]

“Critique your response for usability.” [Output]

Dos and don'ts

- ✓ For complex tasks: “ask me any questions you need for context”
- ✓ Tell it to “be concise” or “use bullet points”
- ✓ Set the tone for responses: professional / informal / funny
- ✗ Don't ask leading questions
- ✗ Don't enter confidential info

Structure a super prompt

“You are a Software Developer. Create 5 email subjects. The email is to inform all the developers about a UX problem with search. The goal is to get them to critique my YouTube video. Subject < 50 characters.”

Role
Result
Context
Intent
Constraint

ssw.com.au/rules/chatgpt-cheat-sheet

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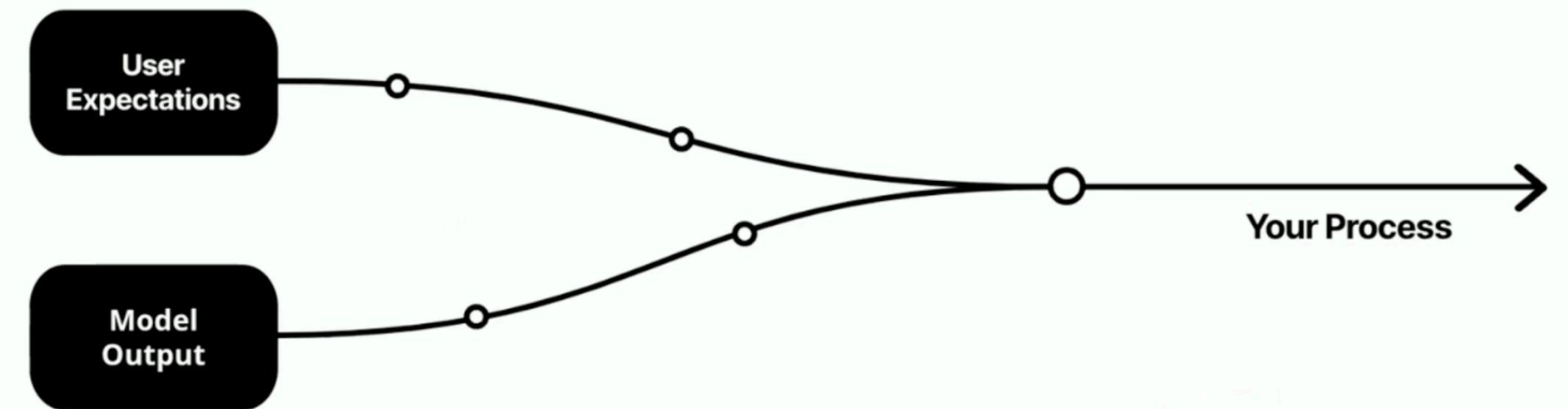
SSW

<https://www.ssw.com.au/rules/static/342f93bb776e37da664f6ccc4e461764/be464/chatgpt-cheat-sheet-bathroom-door.jpg>

Prompt Engineering Primer

- **Prompt engineering** is an iterative process usually
- Can benefit from:
 - **Delimiters** can highlight logical breaks
 - **Step-by-step instructions**
 - **One/few shot learning** (e.g. give it examples of what you want)
 - **Ask LLMs** to help write effective prompts

Iterative Approach

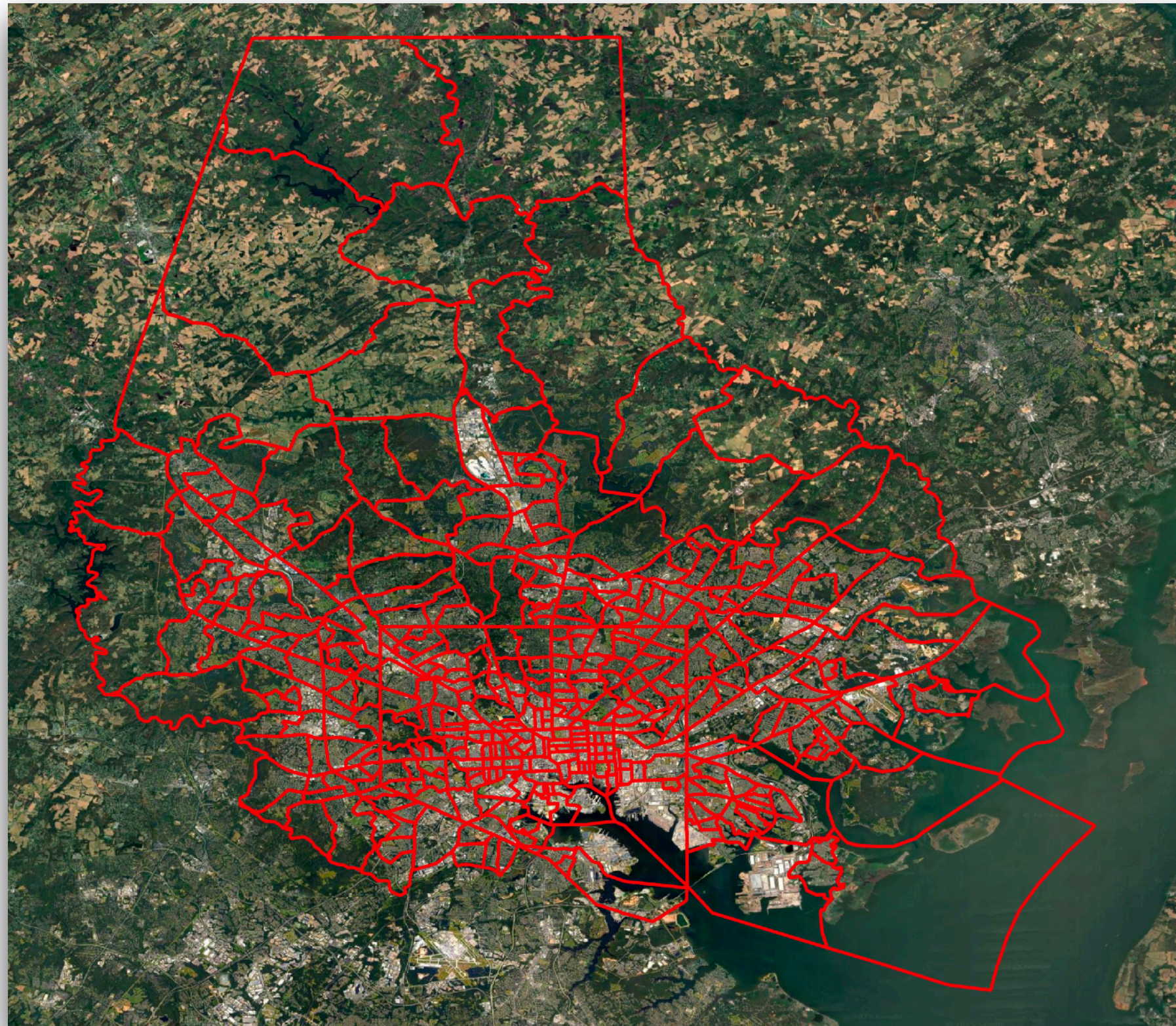


Goals for the Session

- Become comfortable working in **Google Colab** and building a workflow in **Python** even when you don't know Python!!
- Explore the **powerful capabilities** and **common pitfalls** of working with LLMs to code.
- **Have fun!!**

Visual Roadmap for the Session

- **Part 1: Introduction to Google Colab environment and some very basic Python**



Baltimore City and County Census Tracts we will use

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prompts_LLM_coding_tutorial.ipynb

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✓ Analyzing Environmental Justice: Integrating Census and Harmonized Landsat Sentinel-2 (HLS) NDVI Data with LLMs in Google Colab

Overview

This exercise guides environmental and social scientists through the process of analyzing environmental justice issues by combining US Census data with NASA's Harmonized Landsat-Sentinel (HLS) satellite imagery using Large Language Models and rprompt engineering. You'll develop skills in integrating geospatial analysis and socioeconomic datasets using to identify hotspots of social vulnerability.

Learning Objectives

- Learn the basics of Python and using Google Colab
- Integrate socioeconomic and remote sensing datasets
- Perform spatial analysis to identify areas of environmental concern
- Visualize and communicate environmental justice findings
- Become confident in using LLMs and prompt engineering to help with all your coding!!!

Prerequisites

- Basic understanding of coding in some language (R, Matlab, etc)
- Basic understanding of data manipulation, data types, and LLMs
- A great attitude and the eye of the tiger!

✓ Part 1: Brief Introduction to Colab and Setting Up Your Environment and Working Directory

Key Tasks

- Introduction to Google Colab and its features
- Install a suite of packages we will use in our workspace
- Set up our working directory

[] # Install required packages

!pip install pandas geopandas numpy matplotlib seaborn scikit-learn rasterio earthpy rasterstats

✓ PROMPT: "I would like to set my working directory for my notebook"

[] Start coding or generate with AI.

✓ Part 2: Working with Census Data to Create a Social Vulnerability Index (SVI)

In this section, you'll work with a subset of the 2022 Census data from the Baltimore, MD region and create a basic social vulnerability index.

Key Tasks

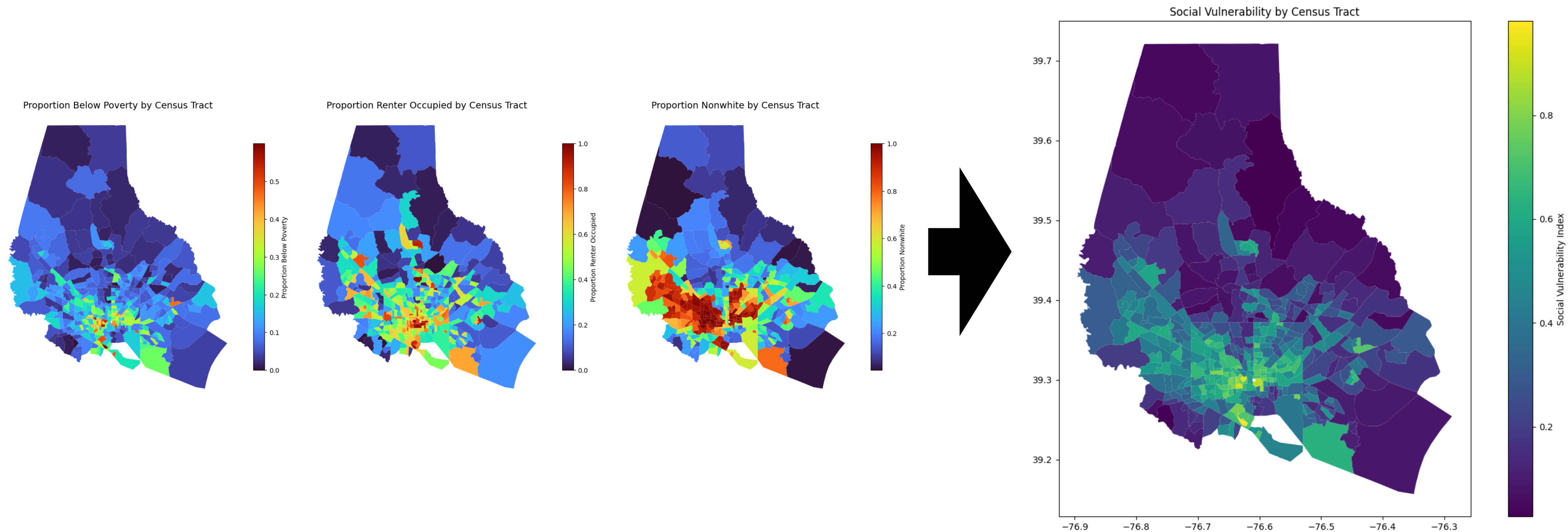
- Read in a .geojson file (essentially a spatial dataframe)
- Learn to investigate variables
- Create and view a simple social vulnerability index (SVI)

✓ PROMPT: "I want to import a .geojson file with census data in it named baltimore_region_acs_2022_epsg4269.geojson from my working directory and get some info about it. I would like the variable to be called gdf"

[] Start coding or generate with AI.

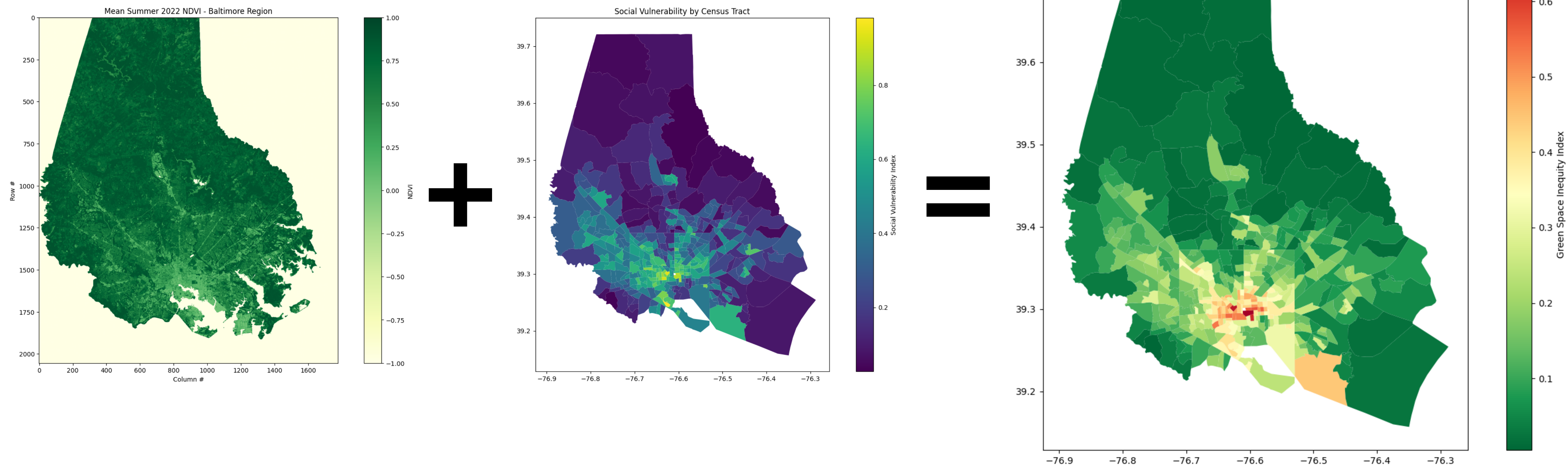
Visual Roadmap for the Session

- **Part 2: Working with Census Data to Create a Social Vulnerability Index (SVI)**



Visual Roadmap for the Session

- Part 3: Integrating Harmonized Landsat Sentinel-2 NDVI Data with Census Data to Illuminate Patterns of Social Vulnerability



Visual Roadmap for the Session

- Part 4: Go Wild with LLM!!



Visual Roadmap for the Session

- **Part 5: Show and Tell: Creating a Dashboard for Environmental Justice Data Exploration Using an LLM**

PROMPT to DALL-E:

Can you make an image that shows a cosmic large language model giving a gift to an environmental scientist?

